

Foundations of Mathematical optimization

[Mathematical optimization]

1.0 Content Block

Computational optimization techniques To solve problems, researchers may use algorithms that terminate in a finite number of steps, or iterative methods that converge to a solution (on some specified class of problems), or heuristics that may provide approximate solutions to some problems (although their iterates need not converge).

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Methods that evaluate Hessians (or approximate Hessians, using finite differences): Newton's method Sequential quadratic programming: A Newton-based method for small-medium scale constrained problems. Some versions can handle large-dimensional problems. Interior point methods: This is a large class of methods for constrained optimization, some of which use only (sub)gradient information and others of which require the evaluation of Hessians. Methods that evaluate gradients, or approximate gradients in some way (or even subgradients): Coordinate descent methods: Algorithms which update a single coordinate in each iteration Conjugate gradient methods: Iterative methods for large problems. (In theory, these methods terminate in a finite number of steps with quadratic objective functions, but this finite termination is not observed in practice on finite-precision computers.) Gradient descent (alternatively, "steepest descent" or "steepest ascent"): A (slow) method of historical and theoretical interest, which has had renewed interest for finding approximate solutions of enormous problems. Subgradient methods: An iterative method for large locally Lipschitz functions using generalized gradients. Following Boris T. Polyak, subgradient-projection methods are similar to conjugate-gradient methods. Bundle method of descent: An iterative method for small-medium-sized problems with locally Lipschitz functions, particularly for convex minimization problems (similar to conjugate gradient methods). Ellipsoid method: An iterative method for small problems with quasiconvex objective functions and of great theoretical interest, particularly in establishing the polynomial time complexity of some combinatorial optimization problems. It has similarities with Quasi-Newton methods. Conditional gradient method (Frank-Wolfe) for approximate minimization of specially structured problems with linear constraints, especially with traffic networks. For general unconstrained problems, this method reduces to the gradient method, which is regarded as obsolete (for almost all problems). Quasi-Newton methods: Iterative methods for medium-large problems (e.g. $N < 1000$). Simultaneous perturbation stochastic approximation (SPSA) method for stochastic optimization; uses random (efficient) gradient approximation. Methods that evaluate only function values: If a problem is continuously differentiable, then gradients can be approximated using finite differences, in which case a gradient-based method can be used. Interpolation methods Pattern search methods, which have better convergence properties than the Nelder-Mead heuristic (with simplices), which is listed below. Mirror descent

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Global convergence More generally, if the objective function is not a quadratic function, then many optimization methods use other methods to ensure that some subsequence of iterations converges to an optimal solution. The first and still popular method for ensuring convergence relies on line searches, which optimize a function along one dimension. A second and increasingly popular method for ensuring convergence uses trust regions. Both line searches and trust regions are used in modern methods of non-differentiable optimization. Usually, a global optimizer is much slower than advanced local optimizers (such as BFGS), so often an efficient global optimizer can be constructed by starting the local optimizer from different starting points.

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Necessary conditions for optimality One of Fermat's theorems states that optima of unconstrained problems are found at stationary points, where the first derivative or the gradient of the objective function is zero (see first derivative test). More generally, they may be found at critical points, where the first derivative or gradient of the objective function is zero or is undefined, or on the boundary of the choice set. An equation (or set of equations) stating that the first derivative(s) equal(s) zero at an interior optimum is called a 'first-order condition' or a set of first-order conditions. Optima of equality-constrained problems can be found by the Lagrange multiplier method. The optima of problems with equality and/or inequality constraints can be found using the 'Karush–Kuhn–Tucker conditions'.

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Calculus of variations is concerned with finding the best way to achieve some goal, such as finding a surface whose boundary is a specific curve, but with the least possible area. Optimal control theory is a generalization of the calculus of variations which introduces control policies. Dynamic programming is the approach to solve the stochastic optimization problem with stochastic, randomness, and unknown model parameters. It studies the case in which the optimization strategy is based on splitting the problem into smaller subproblems. The equation that describes the relationship between these subproblems is called the Bellman equation. Mathematical programming with equilibrium constraints is where the constraints include variational inequalities or complementarities.

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This denotes the minimum value of the objective function $x^2 + 1$, when choosing x from the set of real numbers $\mathbb{R}$. The minimum value in this case is 1, occurring at $x = 0$. Similarly, the notation

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the expression $f(x^*) \leq f(x)$ holds; that is to say, on some region around x^* all of the function values are greater than or equal to the value at that element. Local maxima are defined similarly. While a local minimum is at least as good as any nearby elements, a global minimum is at least as good as every feasible element. Generally, unless the objective function is convex in a minimization problem, there may be several local minima. In a convex problem, if there is a local minimum that is interior (not on the edge of the set of feasible elements), it is also the global minimum, but a nonconvex problem may have more than one local minimum not all of which need be global minima. A large number of algorithms proposed for solving the nonconvex problems – including the majority of commercially available solvers – are not capable of making a distinction between locally optimal solutions and globally optimal solutions, and will treat the former as actual solutions to the original problem. Global optimization is the branch of applied mathematics and numerical analysis that is concerned with the development of deterministic algorithms that are capable of guaranteeing convergence in finite time to the actual optimal solution of a nonconvex problem.

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Control engineering Mathematical optimization is used in much modern controller design. High-level controllers such as model predictive control (MPC) or real-time optimization (RTO) employ mathematical optimization. These algorithms run online and repeatedly determine values for decision variables, such as choke openings in a process plant, by iteratively solving a mathematical optimization problem including constraints and a model of the system to be controlled.